

SmartBite: Food Recognition & Nutrition Estimator

Project in Machine Learning and Data in Operations

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Project Goal

- Context:

- Today, a lot of people behave in a health-conscious manner. They do a lot of sports and pay attention to their eating habits (e.g. calories, fat, etc.).
- To support a healthy lifestyle, we want to provide a tool that detects what you are eating and gives you an overview of the calories.

- Goal

- This project aims to implement an end-to-end machine learning system that recognizes different types of food from images and provides estimated nutritional information for the classified food.

Data

- Data

- Food-101 Dataset (https://data.vision.ee.ethz.ch/cvl/datasets_extra/food-101/)

- Overview

- 101'000 images of food
 - 101 classes (different food types)
 - 1000 images per class (max. 512 x 512 pixels)
 - 750 train images per class
250 test images per class

- Food and nutrition data API

- <https://platform.fatsecret.com/api-demo>



Example image: Crème brûlée

Important to consider

- We aim to:
 - Version code to always roll back to a working version if a new code breaks down, and to collaborate together.
 - Track the model training to have an overview and compare the different versions.
 - Track the data to always know which version was used for which model training.
 - Create a dashboard for an easy, interactive way to get predictions.
 - Develop an API to classify food images and provide users with corresponding nutrition facts.

Tools

Tool	Purpose	Why we use this tool
PyTorch	Model training	Flexible, powerful DL framework, pretrained models
Weights & Biases (W&B)	Experiment tracking	Visual logging, interactive, real-time, hyperparameter sweeps
Streamlit	Interactive UI for users to interact with the model	Quick to build dashboards, great for simple visual prototypes of ML apps
FastAPI	Backend API for predictions	Lightweight, fast and easy to integrate with ML
GitHub	Code version control, collaboration, and CI/CD pipeline	Decentralized development, code versioning, CI/CD-ready
Docker	Containerization of our application components (FastAPI + Streamlit)	Ensures consistent and reproducible environments. Simplifies deployment and dependency management.
DVC and DigitalOcean	Data/model versioning	Ensures reproducibility and consistent dataset tracking